Code No: 136CF **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, July - 2023
INTRODUCTION TO ANALYTICS
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a)	Define outlier.	[2]
b)	Discuss about conditional statements in R.	[3]
c)	State central limit theorem.	[2]
d)	Give the importance and properties of normal distribution.	[3]
e)	What is xlsx?	[2]
f)	Differentiate SQL and NoSQL databases.	[3]
g)	Define regression.	[2]
h)	What is Pearson Correlation?	[3]
i)	What is smart utility system?	[2]
j)	What are the applications of business intelligence?	[3]

PART – B

(50 Marks)

2.a)	Describe different data types in R.	
b)	Explain for loop in R programming and how it is useful in data frame.	[5+5]
	OR	
3.a)	Create a matrix using rbind() and cbind() and illustrate with examples.	
b)	Write the importance of work management and prioritization.	[5+5]
4.a)	Describe summery(), fix(), str(), aggregate() and name() functions.	
b)	What is random variable? Explain its categories.	[5+5]
	OR	
5.a)	Explain the pros and cons of team work in analytics.	
b)	Explain Normal probability distribution function in detail.	[5+5]
6.a)	Explain different packages to access Excel sheet from R.	
b)	Write the steps to connect R to NoSQL data base.	[5+5]
	OR	
7.a)	How will you execute R code from an Excel spreadsheet?	
b)	List and explain the classifications of NoSQL databases based on data model.	[5+5]

8.a) What is OLS regression? Write the assumptions of OLS regression.
b) What is ANOVA model? Explain it briefly. [5+5]

OR

9.a) What is residual plot and explain how to find residuals.
b) Give a note on Heteroscedasticity. [5+5]

10.a) Discuss the stages of engineering design process.
b) Compare business analytics and organization decision-making processes. [5+5]

OR

11.a) What is SMART? Explain in detail.
b) Explain the components of Business intelligence. [5+5]

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